UCONN FALL FEST 2026 · IBM QISKIT × qBRAID CHALLENGE

# Instant Insanity on a Quantum Computer

Constrained QAOA with W-states and XY mixers, checked against brute force at every step


## Team Phase Flip

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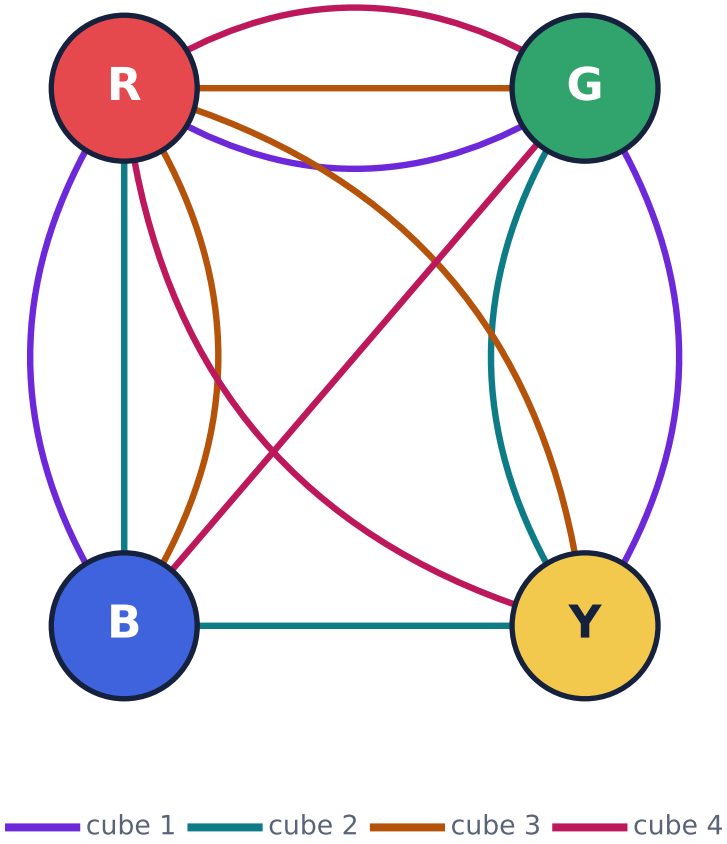

Levels reached: Warm-up ✓ Core (4 cubes) ✓ Bonus (N = 5 to 8 cubes) ✓

github.com/scarthur/IBM-Qiskit-Team-Phase-Flip

# The puzzle: no colour may repeat on any side

- Each cube has 3 pairs of opposite faces in 4 colours (R, G, B, Y).
- Stack all 4 cubes: front, back, left and right must each show every colour exactly once.
- Each cube has 24 orientations: $24^4 = 331{,}776$ towers to try, and only 8 of them work (one tower, 8 symmetric views).
- Graph view: one node per colour, one edge per opposite pair.
- One axis (e.g. front/back) = one edge per cube, every colour touched twice: a 2-factor.
- Solution = two 2-factors that never use the same pair of a cube. Of $6^4 = 1{,}296$ graph cases, exactly 2 solve it.

Colour graph of our 4 cubes



# Encoding: one-hot picks of opposite-face pairs

- x[c][p] = 1 if cube c puts pair p on front/back;  y[c][p] for left/right.
- 6 qubits per cube → 24 qubits for the full tower (qubit 6c+p = x, 6c+3+p = y).
- Warm-up and bonus: front/back only, 3 qubits per cube (3N qubits).
- Cost (QUBO, zero exactly for a solved tower):

$$C = \sum_{\text{axis } k}\sum (d_k - 2)^2 \; + \; \mu \sum_{c,p} x_{cp}\, y_{cp} \; + \; \lambda \sum_{c} \Big[ \big(\sum_{p} x_{cp} - 1\big)^2 + \big(\sum_{p} y_{cp} - 1\big)^2 \Big]$$

- (d_k − 2)²: every colour k must appear exactly twice on each axis (the 2-factor rule).
- μ·x·y: a pair can't serve both axes.  λ: one pair per axis. We use λ = μ = 2.
- Mapped to Pauli Z / ZZ terms with x = (1 − Z)/2. In the circuit the one-hot λ term is dropped:

  the W-state + XY mixer never leaves the one-hot subspace.

# Algorithm: constrained QAOA over legal picks only

Each cube starts in a W-state $(|100\rangle+|010\rangle+|001\rangle)/\sqrt{3}$ and is mixed with XX+YY gates, so every shot is a valid one-hot pick.

p = 2 layers; angles found with COBYLA on a 12-qubit statevector (local Aer, SEED = 7).

### Method A · 24 qubits

Full tower in one circuit.

Double W-state per cube (front/back + left/right).

Cost: degree penalties + μ·x·y.

Angles transferred from the 12-qubit warm-up (not re-optimised at 24 qubits).

### Method B · hybrid 12q + 4q

Stage 1 (quantum): 12-qubit W+XY QAOA samples front/back 2-factors.

Stage 2 (quantum): for a sampled 2-factor, a 4-qubit QAOA picks one of the 2 remaining pairs per cube for left/right.

### Bonus · N cubes, N colours

Stage 1 (quantum): 3N-qubit W+XY QAOA, same p = 2 angles.

Stage 2 (classical): pair two sampled 2-factors that share no edge within a cube; fall back to the 4-qubit-style QAOA if none.

We say which part is quantum.

# Classical checks before claims

✓ **Hamiltonian = classical cost on all 1,296 valid choices**

max |H − cost| = 0.0e+00 (assert in notebook)

✓ **Exactly 2 zero-energy states**

the tower and its quarter turn, as brute force predicts

✓ **Both ground states decode to a real tower**

solves(tower_from(...)) is True for each

✓ **Decoder respects Qiskit bit order**

qubit 0 = rightmost character; tested on the known answer

✓ **Every reported tower re-checked**

independent solves() on every decoded shot we count

✓ **Bonus puzzles have exactly one tower**

seeds {5:16, 6:86, 7:68, 8:4}: brute force finds one edge-disjoint pair of 2-factors

Verified tower (from both methods)

| | Front | Back | Left | Right |
|---|---|---|---|---|
| cube 1 | G | Y | B | R |
| cube 2 | Y | B | Y | G |
| cube 3 | B | R | R | Y |
| cube 4 | R | G | G | B |

Each column shows R, G, B, Y once

Honest note: brute force solves the 4-cube puzzle in under a second.

# Results: 4-cube puzzle (local AerSimulator, SEED = 7)

| Experiment | Qubits | CX | Depth | Shots | Measured | Random guess | Gain |
|---|---|---|---|---|---|---|---|
| Warm-up: any front/back 2-factor | 12 | 300 | 218 | 1000 | **28.2%** | 4.94% (4/81) | 5.7× |
| Method A: full solved tower | 24 | 648 | 224 | 2000 | **1.60%** | 0.0305% (2/6561) | 52× |
| Method B stage 1: 2-factor | 12 | 300 | 218 | 1000 | **28.2%** | 4.94% (4/81) | 5.7× |
| Method B stage 2: tower \| FB | 4 | 12 | 29 | 1000 | **26.2%** | 6.25% (1/16) | 4.2× |

- Every counted shot passes solves(); random baselines are uniform over the same encoding's legal (one-hot) strings.
- Method A, over all $2^{24}$ bitstrings: 2 / 16.8 M = 0.000012%. The constraint-preserving mixer does most of the work.
- Method B stage 2 is conditional on a good front/back 2-factor: then exactly 1 of 16 left/right picks solves.
- Its 4-qubit circuit is tiny (12 CX, depth 29).

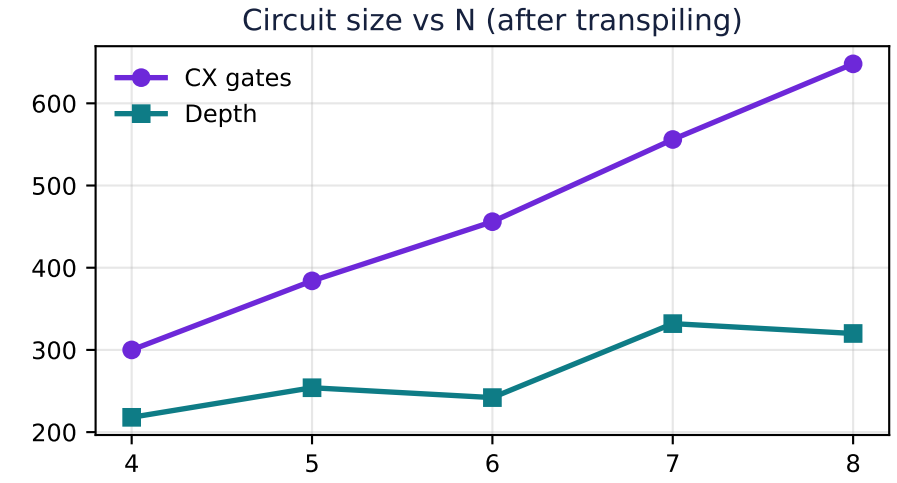
**What it tells us: QAOA puts far more weight on answers than chance does, but 1.6% at 24 qubits shows the full one-circuit encoding is hard at p = 2; splitting the problem (Method B) works much better.**

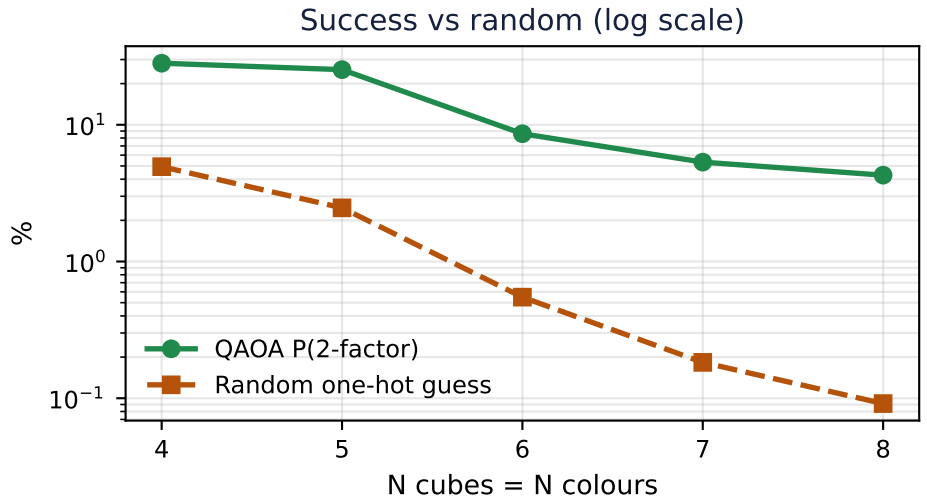
SCALING

# Bonus: the same circuit scales to N = 8

| N (seed) | Qubits | CX | Depth | P(2-fac.) | Random | Gain | Tower |
|---|---|---|---|---|---|---|---|
| 4 (CUBES) | 12 | 300 | 218 | **28.2%** | 4.94% | 5.7× | ✓ |
| 5 (16) | 15 | 384 | 254 | **25.3%** | 2.47% | 10× | ✓ |
| 6 (86) | 18 | 456 | 242 | **8.6%** | 0.55% | 16× | ✓ |
| 7 (68) | 21 | 556 | 332 | **5.3%** | 0.18% | 29× | ✓ |
| 8 (4) | 24 | 648 | 320 | **4.3%** | 0.091% | 47× | ✓ |

- Random = number of 2-factors / $3^N$ (brute-force count).
- CX count grows about linearly: 300 → 648.
- Success falls 28% → 4.3%, but its lead over random guessing grows from 5.7× to 47×.
- A solves()-checked tower was found for every N by classically pairing sampled 2-factors.



TAKEAWAYS

# What we learned, and what's next

## What we can and can't claim

- ✓ The model is exact: H matches brute force on all 1,296 cases.
- ✓ Constrained QAOA beats random guessing in every experiment (5.7× to 52×).
- ✓ Every tower we report passes solves().

- ✗ No quantum speedup: brute force is instant here and at N = 8 ($6^8 \approx 1.7$ M cases).
- ✗ Results are from local AerSimulator (seed 7). The SV1 path is wired in (USE_SV1 = True).
- ✗ 24-qubit run reused 12-qubit angles: 1.6%.

## One thing we'd try next

Re-optimise Method A at 24 qubits with a CVaR objective (score only the best shots), warm-started from our p = 2 angles, then grow to p = 3 layer by layer.

Then: a mixer over each cube's 6 valid (front/back, left/right) choices, so the "pair can't serve both axes" rule is enforced by the circuit, not a penalty.

Run the final circuits on SV1 and compare with hardware using earned credits.